Politecnico di Torino

Master's Degree in Mathematical Engineering

Material for Thesis

c– Data Pre-processing Pipeline
Intra and Inter Analysis

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Contents

Abstract

MANCA L'ABSTRACT.

1 Intra-Dataset Pre-processing

Data pre-processing represents a foundational stage in DNA methylation analysis, particularly in cancer epigenomics, where technical artefacts, probe-design biases, and unwanted sources of variation can obscure or distort genuine biological signals. Before performing feature selection, or machine-learning models, it is essential to ensure that the methylation matrices used for statistical modelling are accurate, and free from unreliable probes or batch-associated effects.

In the context of breast cancer, where early epigenetic alterations may be subtle and highly localized, careful pre-processing is crucial to avoid confounding Normal–Adjacent differences with technical noise. Several well-established sources of unreliability—cross-reactive probes, SNP-affected loci, non-CpG targets, Infinium Type I/II probe-design bias, and batch effects—must be addressed through a systematic and reproducible workflow.

The primary goal of this section is therefore to construct, for each dataset, a high-quality and biologically coherent methylation matrix suitable for downstream analyses. To this end, all datasets undergo the same structured pipeline: initial alignment and quality checks; removal of unreliable CpGs through literature-based technical filtering; identification of invariant CpGs that carry no discriminative information along the Normal–Adjacent axis; conversion from β-values to M-values for statistical modelling; and correction of batch effects.

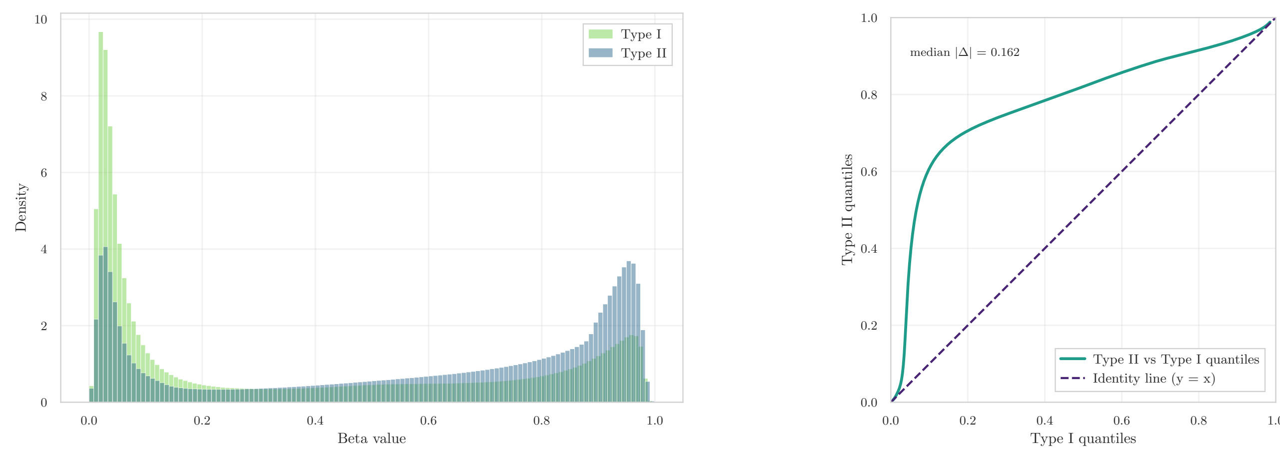
These steps ensure methodological consistency across the three cohorts (GSE69914 [1.1], GSE225845 [1.2], GSE287331 [1.3]) and provide a unified, denoised, and biologically meaningful set of features to be used in the subsequent analyses of early epigenetic alterations.

1.1 Dataset GSE69914

Step 0 – Initial Pre-processing The initial pre-processing stage verifies the structural integrity of the dataset. The β-value matrix and its corresponding phenotype table were imported from their Parquet representations. Both tables contained ✓ 397 samples, and the β-matrix included ✓ 485,512 CpGs, confirming full dataset availability at load time.

Sample identifiers in the phenotype table were inspected to ensure uniqueness and correct metadata structure. No duplicated `id_tissue` entries were detected ✓ (0 duplicates), and all required fields (`id_tissue`, `label`) were present ✓ (0 missing fields), indicating that the phenotype annotation is complete and coherent. Alignment between the phenotype table and the β-matrix was assessed by intersecting sample IDs. All samples were shared across matrices, yielding ✓ perfect correspondence (397/397 matched).

A probe-level and sample-level missingness check was then performed to verify the completeness of the methylation matrix. The entire dataset was found to be fully observed, with ✓ 0 missing CpG values and ✓ 0 missing sample entries. This confirms that no early NaN filtering or imputation is required for this dataset.



(a) β-value distribution by Infinium probe design (Type I vs. Type II). The near-overlapping shapes indicate effective correction of probe-type bias.

(b) Q–Q plot comparing quantiles of Type II vs. Type I probes. The alignment to the diagonal ($y = x$) confirms balanced signal distributions after normalization.

Figure 1: Diagnostic evaluation of Infinium Type I/II probe bias correction in the GSE69914 dataset. The consistent β-value and quantile distributions demonstrate the effectiveness of the BMIQ normalization previously applied.

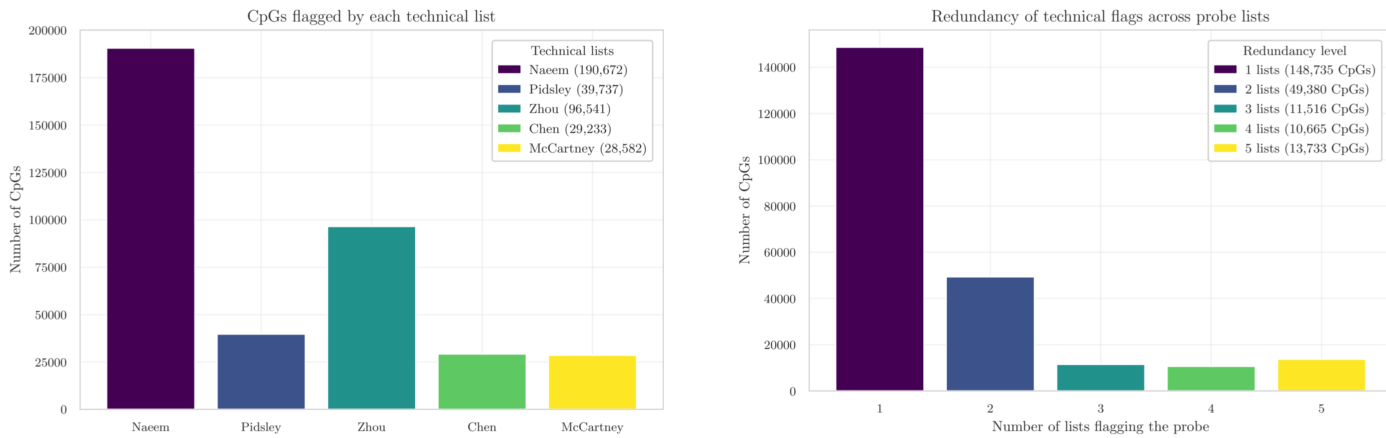
Step 1 – Infinium Type I/II Bias Diagnostics Raw methylation intensity data (*IDAT* files) were processed by the original authors using the `minfi` pipeline and `BMIQ` normalization, as reported in [1]. A well-known characteristic of Illumina Infinium microarrays is the **systematic technical bias between Type I and Type II probes**. This bias has been extensively documented in the literature, most notably by Maksimovic *et al.* [2], who showed that the two probe chemistries produce inherently different β-value distributions and therefore require appropriate probe-type normalization. Such differences, if uncorrected, can propagate into downstream statistical analyses.

To independently verify that this bias had been successfully mitigated in the present dataset, β-values were stratified by probe design (Type I vs. Type II) using the official Illumina 450K manifest from Bioconductor [3]. Following the diagnostic procedure of Teschendorff *et al.* [4], two complementary assessments were performed:

1. **Density comparison.** As shown in Fig. 1a, the β-value distributions of Type I and Type II probes substantially overlap, indicating the absence of the multi-modal distortion typical of unnormalized data.

2. **Quantile–quantile alignment.** The Q–Q plot in Fig. 1b exhibits quite a diagonal trend for Type II vs. Type I quantiles, which is the expected signature of balanced probe-type scaling after normalization.

The resulting pattern closely matches the characteristic behaviour described by Teschendorff *et al.* [4] and Wang *et al.* [5], while being fully consistent with the probe-type discrepancies originally analysed by Maksimovic *et al.* [2]. ✓ Together, these diagnostics confirm that the dataset has undergone effective Type I/II probe bias correction.



(a) CpGs flagged by each technical filtering resource (Naeem, Pidsley, Zhou, Chen, McCartney).

(b) Redundancy distribution of probes flagged by 1–5 lists.

Figure 2: Summary of technical filtering. **(a)** Intersection between the dataset and each external probe-filtering resource, showing the number of CpGs flagged by Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis indicating how many probes were simultaneously flagged by 1 to 5 independent resources, quantifying the concordance among technical annotations.

Step 2 – Technical Filtering Technical filtering aims to remove unreliable or biologically confounded probes prior to statistical modelling. This stage reduces noise, improves downstream reproducibility, and ensures that only high-confidence CpG loci are retained for analysis (Fig. 2).

Exclusion of technical probe sets. A collection of curated probe resources was used to exclude loci known to exhibit technical artefacts (see Appendix A for details). These include probes affected by common genetic variants at the CpG or single-base extension site [6], [7], probes with demonstrated multi-mapping or off-target hybridisation [8], [9], and probes flagged by the `MASK_*` reliability annotations [6]. Additional quality control criteria follow the hierarchical filtering strategy of Naeem et al. [10], which targets multi-mapping loci, repeat structures, and disruptive SNPs.

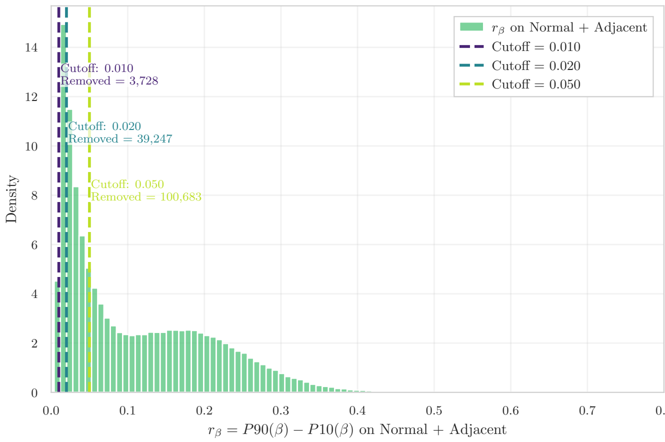
The overlaps between these resources and the β-matrix were substantial (Fig. 2a). A total of ✓ 190,672 probes were flagged by Naeem et al., ✓ 39,737 by Pidsley et al., ✓ 29,233 by Chen et al., ✓ 96,541 by the Zhou `MASK_*` fields, and ✓ 28,582 by McCartney et al. These lists are known to partially overlap—particularly those focused on multi-mapping and SNP-associated unreliability—and their union resulted in the removal of ✓ 225,426 unique CpGs. After this operation, ✓ 260,086 CpGs remained.

Annotation-based filtering (Illumina manifest). To ensure consistency with the manufacturer's official annotation, the HumanMethylation450 v1.2 manifest [11] was used to validate probe identity and genomic mapping. Through this annotation, probes lacking valid chromosome information, those with undefined genomic coordinates, and those targeting non-CpG contexts (IDs beginning with "`ch`") were identified and removed. This step aligns the working feature set with Illumina's reference genome definition and follows the recommendations of Zhou et al. [6] and Pidsley et al. [7]. A total of ✓ 875 non-CpG probes were excluded, reducing the feature space to ✓ 259,211 CpGs.

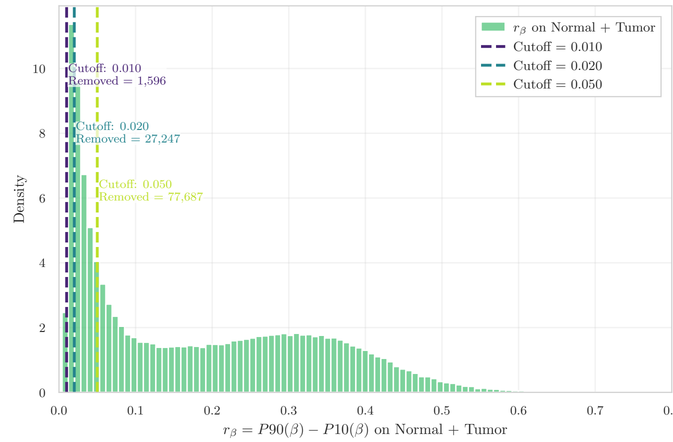
Removal of sex-chromosome probes (X/Y). To avoid sex-related confounding effects and to maintain cross-dataset consistency, all probes located on chromosomes X and Y were removed based on manifest annotations. This choice is consistent with established preprocessing workflows for Illumina 450K data, where probes on sex chromosomes are excluded when male and female samples are mixed, as recommended by Maksimovic et al. [2]. Moreover, sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and joint normalization of autosomal and sex-chromosome probes has been shown to introduce artificial sex-related bias into thousands of autosomal CpGs [12]. Removing X/Y probes therefore provides a conservative and robust strategy when sex-specific methylation is not under investigation. This filtering step eliminated ✓ 7,728 CpGs, producing a final autosomal working set of ✓ 251,483 CpGs.

Overall, the initial set of ✓ 485,512 CpGs was reduced to ✓ 251,483 CpGs after applying all technical lists, manifest-based validation, and sex-chromosome exclusion. In total, ✓ 234,029 unique CpGs were removed across all criteria. A per-probe summary file `removed_cpgs_all_filters_summary_gse69914.csv` records, for each excluded probe, which resources contributed to its removal, ensuring full reproducibility of the filtering pipeline.

Step 3 – Invariance Filtering on Normal+Adjacent A stage of feature-level pre-processing was applied to remove CpG loci that exhibit minimal biological variability in **normal breast tissue**. This strategy follows the empirically driven framework proposed by Edgar et al. [13] for detecting non-variable CpGs across tissues using the β-value range

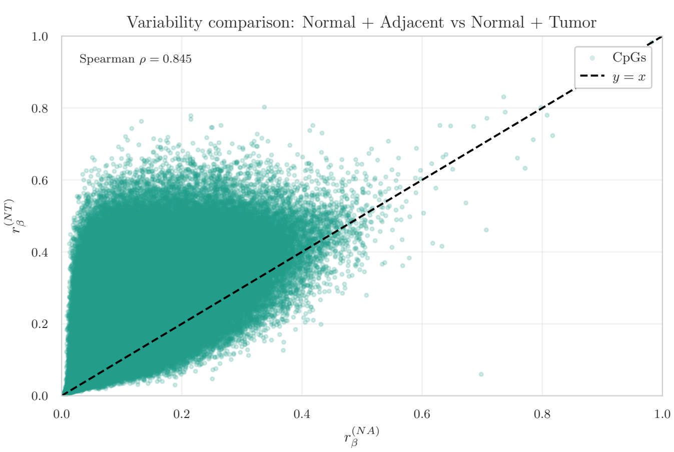


(a) r_β distribution on Normal+Adjacent samples, with thresholds at 0.01, 0.02, and 0.05.

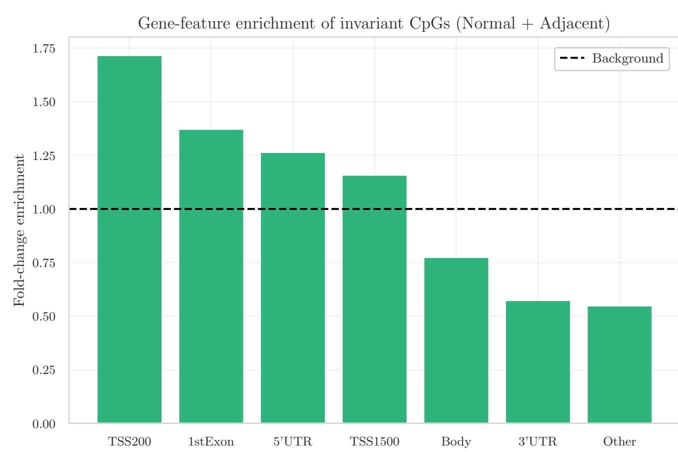


(b) r_β distribution on Normal+Tumor samples, with thresholds at 0.01, 0.02, and 0.05.

Figure 3: **Variability range analysis**. The Normal+Adjacent subset displays a large block of CpGs with extremely low variability ($r_\beta < 0.05$), supporting the use of invariance filtering to reduce biologically uninformative loci.



(a) Comparison of CpG variability between N+A and N+T. Breakdown loci diverge from the identity line.



(b) Genomic enrichment of invariant N+A CpGs across gene features.

Figure 4: **Variability concordance and genomic context of invariant CpGs**. Stable loci dominate the diagonal, while breakdown loci reflect tumour-induced methylation variability. Promoter regions show the strongest enrichment of invariant CpGs.

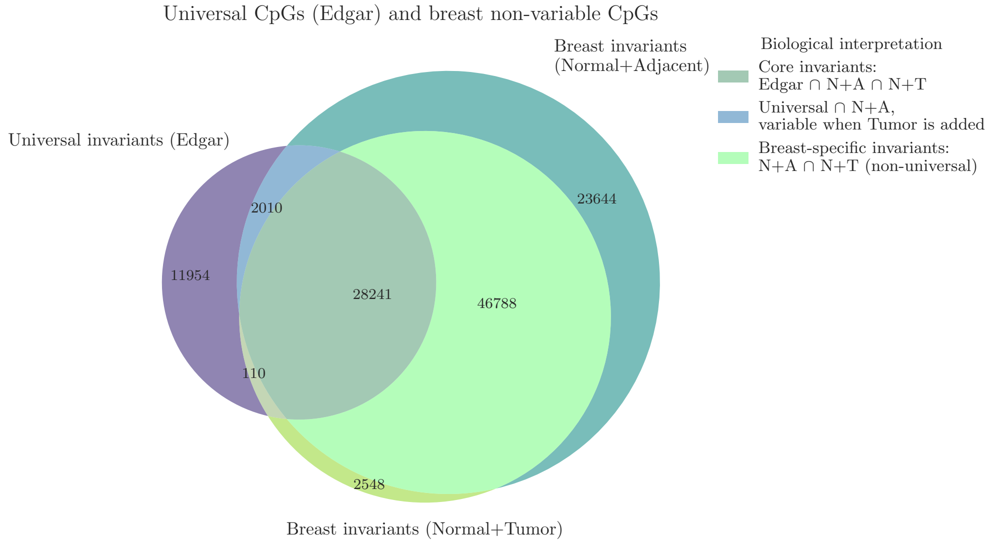


Figure 5: **Intersection between universal invariant CpGs (Edgar) and breast invariants**. Core invariants (Edgar ∩ N+A ∩ N+T) represent strongly conserved loci, whereas "breakdown loci" highlight CpGs losing stability specifically in tumour tissue.

statistic

$$r_\beta = P90(\beta) - P10(\beta),$$

computed here separately on **Normal+Adjacent** (N+A) and **Normal+Tumor** (N+T) subsets.

For the N+A group, r_β was computed for all ✓ 251,483 autosomal CpGs. The distribution of variability revealed that a substantial proportion of CpGs display extremely narrow dynamic range (Fig. 3). Following Edgar et al.'s recommended threshold of $r_\beta < 0.05$, a total of ✓ 100,683 CpGs (40.0%) were classified as **non-variable** and removed from subsequent modelling, leaving ✓ 150,800 CpGs as the **post-invariance feature set** used to construct the matrix `beta_final_preFS` (✓ 397 samples × 150,800 CpGs).

A parallel computation on the N+T subset showed reduced invariance: ✓ 77,687 CpGs (30.9%) were non-variable under the same threshold. Comparing the two invariance sets revealed three biologically informative categories:

- **Stable invariants** (✓ 75,029 CpGs): loci consistently non-variable in both N+A and N+T;
- **Breakdown loci** (✓ 25,654 CpGs): CpGs variable only when tumour samples are introduced, indicating possible *tumour-driven epigenetic disruption*;
- **NT-only invariants** (✓ 2,658 CpGs): CpGs non-variable in N+T but variable in normal breast tissue.

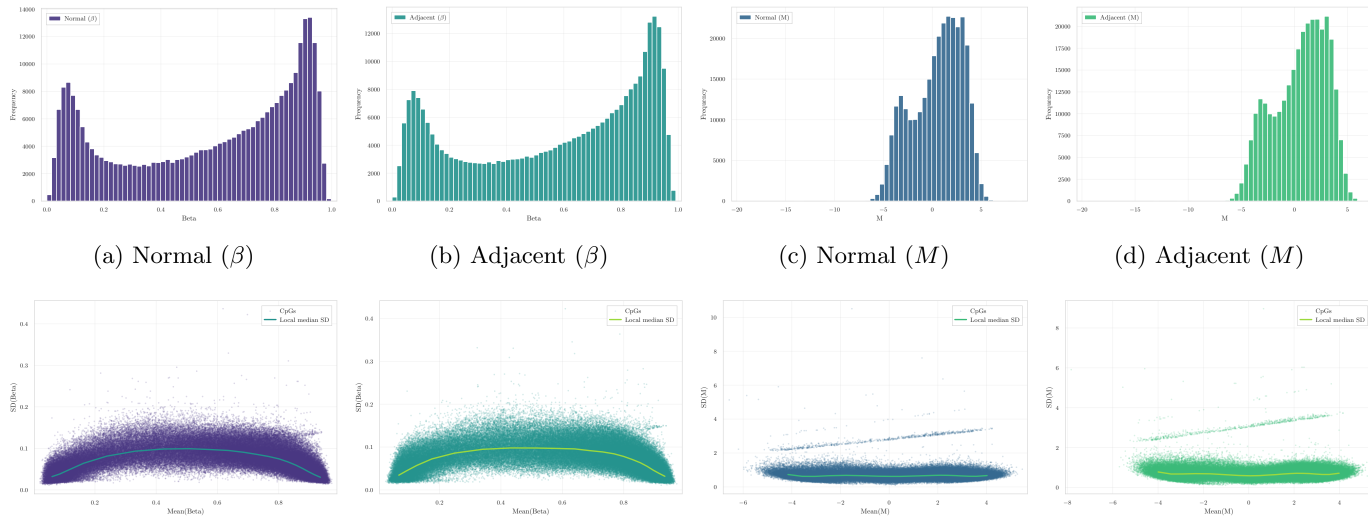
To assess whether variability patterns in N+A were consistent with N+T, a scatter comparison between $r_\beta^{(NA)}$ and $r_\beta^{(NT)}$ was generated (Fig. 4). The two measures exhibited strong concordance (✓ Spearman $\rho = 0.845$), although a clear band of CpGs deviated from the diagonal, corresponding precisely to the **breakdown loci** defined above.

Further annotation-based analysis showed that invariants are not uniformly distributed across genomic elements. Invariant CpGs were enriched in promoter-proximal regions—particularly `TSS200` and `1stExon`—while gene bodies and 3'UTRs were underrepresented (Fig. 4b). This pattern mirrors the findings of Edgar et al. [13], who reported that invariant CpGs preferentially occupy regulatory elements with constitutively stable methylation profiles.

Invariants in breast tissue were compared with the three tissue invariance sets published by Edgar et al. (blood, buccal, placenta). Their intersection yielded ✓ 42,315 universal CpGs shared across all tissues. Among these, ✓ 30,251 remained invariant in Normal+Adjacent breast, and ✓ 28,351 remained invariant even after including tumour samples. A biologically relevant subset of ✓ 2,010 CpGs belonged to the breakdown category, suggesting possible tissue-specific loss of epigenetic stability in tumourigenesis. Furthermore, more detailed comparisons between Edgar et al.'s invariant CpGs and the subset invariant in normal breast tissue are presented in the Inter-Dataset Analysis chapter (see Section 2).

Step 4 – β to M Conversion Illumina Infinium methylation arrays measure methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) signal intensities per CpG. The processed dataset provides only β-values,

$$\beta = \frac{y^{(M)}}{y^{(M)} + y^{(U)}},$$



(a) Normal (β) (b) Adjacent (β) (c) Normal (M) (d) Adjacent (M)

(e) Mean–SD Normal (β) (f) Mean–SD Adjacent (β) (g) Mean–SD Normal (M) (h) Mean–SD Adjacent (M)

Figure 6: **Distributional comparison between β and M quantifications in Normal and Normal-adjacent tissues.** Top row: β exhibits a bounded, U-shaped distribution, whereas M shows an approximately symmetric profile. Bottom row: The substantial mean-dependent variance in β is dramatically reduced after transformation, confirming that M-values provide a more homoscedastic scale for statistical modeling. All panels computed on the working set of ✓ 150,800 CpGs after complete technical and invariance filtering.

a bounded proportion in $[0, 1]$ characterised by strong mean–variance dependence. Following the foundational analysis of Du *et al.* [14], all β-values were transformed into M-values using the logit relation

$$M = \log_2\left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon}\right), \qquad \varepsilon = 10^{-6},$$

which yields an approximately homoscedastic scale more suitable for linear modelling, empirical Bayes moderation, ComBat batch correction, and regression-based inference.

Importantly, the adoption of M-values is not only theoretically motivated but also widely implemented in practice. Several high-impact studies routinely perform statistical modelling in M-space while retaining β-values solely for biological interpretation and effect-size visualisation. Examples include the differential methylation analysis of purified human blood cell types by Reinius *et al.* [15], the methodological framework underlying the `minfi` pipeline described by Triche *et al.* [16], and oncological applications such as the tumour hypoxia study by Thienpont *et al.* [17], where all β-values were explicitly transformed to M-values prior to statistical testing. These examples illustrate that M-values have become a de facto standard for inferential analyses in Illumina 450K/EPIC workflows.

✓ The conversion was applied to the complete post-filtering matrix (150,800 CpGs across 397 samples).

Distributional consequences of the transformation. Fig. 6 illustrates in detail how the $\beta \to M$ mapping reshapes the empirical distribution of methylation. Normal and Adjacent tissues display the expected U-shaped profiles in β-space, reflecting biological accumulation near fully unmethylated or fully methylated states. After logit transformation, the corresponding M-value distributions become substantially more symmetric and comparable across groups, indicating that the transformation regularises the scale without distorting biological structure.

The mean–SD relationships further highlight this effect. In β-space, variance is strongly mean-dependent and inflates near the boundaries, whereas M-values exhibit markedly flatter SD profiles. This variance-stabilising behaviour is consistent with the theoretical findings of Du *et al.* [14] and with the practical motivations underlying the adoption of M-values in modern methylation studies.

Step 5 – Batch Effect Correction

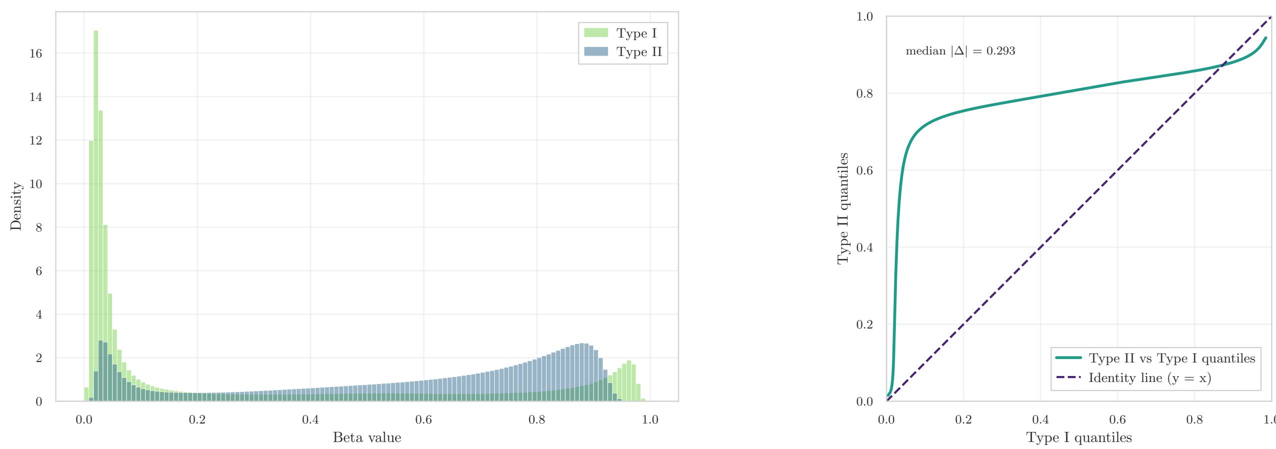
Step 6 – Final Outputs

1.2 Dataset GSE225845

Step 0 – Initial Pre-processing The β-value matrix was successfully imported from its Parquet representation, yielding ✓ 477 samples and ✓ 750,426 CpGs, reflecting the high dimensionality of the Illumina EPIC platform. The corresponding phenotype table contained the same number of samples ✓ (477 entries) and provided all required metadata fields with no inconsistencies.

Sample identifiers were inspected to ensure uniqueness and correct formatting. No duplicated `id_tissue` values were detected, and all metadata columns required for downstream analysis were present ✓ (0 missing fields). Alignment between phenotype records and the methylation matrix was confirmed by intersecting sample IDs, producing ✓ perfect correspondence (477/477 matched).

A probe-level and sample-level missingness check verified the completeness of the methylation data. The dataset was fully observed, with ✓ 0 missing CpG values and ✓ 0 missing sample entries. Accordingly, no early NaN filtering or imputation was required before proceeding to probe-type diagnostics and technical filtering.



(a) β-value distributions for Infinium Type I and Type II probes. The two probe families exhibit markedly different empirical shapes, indicating residual probe-design bias.

(b) Q–Q comparison of Type II vs. Type I β-value quantiles. The substantial deviation from the identity line ($y = x$), with median absolute deviation ✓ 0.2933, indicates incomplete probe-type alignment.

Figure 7: Diagnostic evaluation of Infinium Type I/II probe-design bias. Both the density distributions and the Q–Q pattern reveal that the two probe chemistries retain systematically different scaling behaviour, indicating that residual Type I/II bias persists in the dataset.

Step 1 – Infinium Type I/II Bias Diagnostics All probes were stratified, this yielded ✓ 119,760 Type I and ✓ 627,842 Type II probes.

The density representation of the two probe families (Fig. 7a) reveals a marked discrepancy between their empirical β-value distributions. Type I probes exhibit a highly concentrated pattern near the unmethylated and fully methylated boundaries, whereas Type II probes display a broader and right-shifted distribution. Such divergence indicates that the two probe chemistries retain systematically different scaling behaviour.

This finding is reinforced by the quantile–quantile comparison in Fig. 7b. Type II quantiles deviate substantially from the identity line, with a ✓ median absolute deviation of 0.2933, demonstrating that the expected alignment between the two probe types is not present. The curvature of the Q–Q profile confirms the presence of a residual probe-design imbalance that has not been fully corrected by preprocessing.

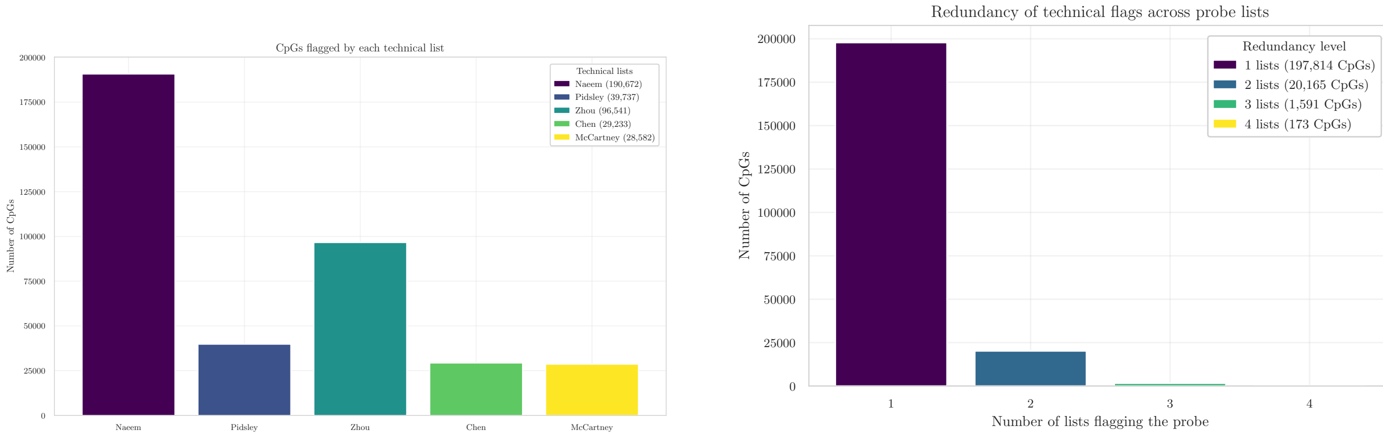
Step 2 – Technical Filtering Beginning from an initial set of ✓ 750,426 CpGs, the intersection with the external probe-filter lists showed substantial overlap across multiple sources.

Exclusion of technical probe sets. The curated technical resources flagged a substantial fraction of probes as potentially unreliable (Fig. 2a): ✓ 190,672 CpGs were flagged by Naeem, ✓ 39,737 by Pidsley, ✓ 96,541 by the Zhou fields, ✓ 29,233 by Chen, and ✓ 28,582 by McCartney. The union of all technical lists resulted in the removal of ✓ 204,899 unique CpGs, leaving ✓ 545,527 CpGs for subsequent steps.

Annotation-based filtering (Illumina manifest). Using the EPIC v1.0 manifest, all probes lacking valid genomic coordinates or targeting non-CpG contexts were removed. This step eliminated ✓ 773 non-CpG loci, reducing the working set to ✓ 544,754 CpGs.

Removal of sex-chromosome probes (X/Y). Probes mapped to chromosomes X and Y were excluded. A total of ✓ 14,071 CpGs were removed, producing a filtered autosomal set of ✓ 530,683 CpGs.

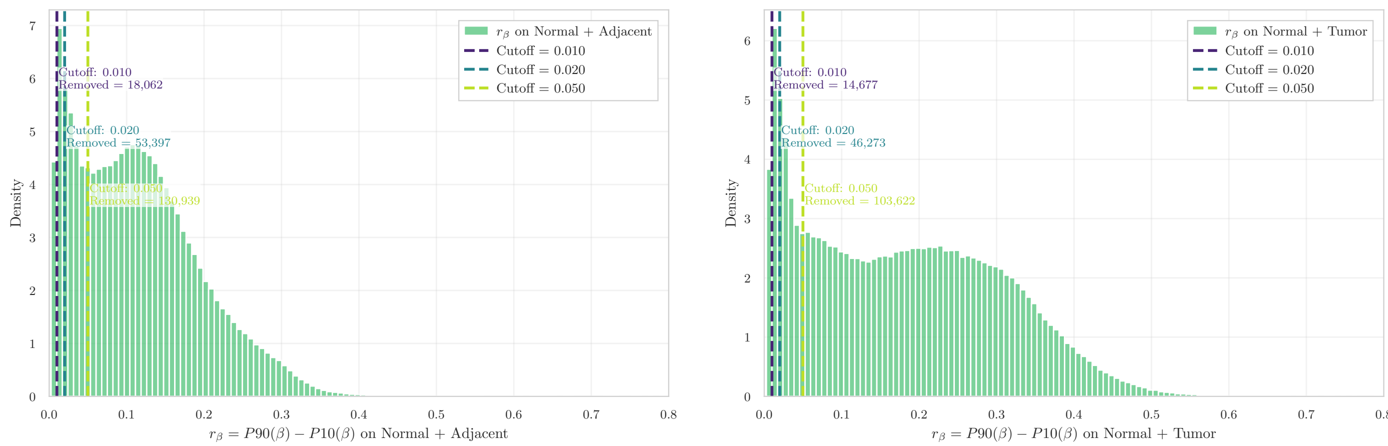
Across all steps, ✓ 219,743 unique CpGs were removed from the initial ✓ 750,426, yielding a final feature set of ✓ 530,683 CpGs. A detailed per-probe summary of the filtering decisions is provided in `removed_cpgs_all_filters_summary.csv`.



(a) Number of CpGs flagged by each technical list.

(b) Redundancy distribution: CpGs flagged by 1–4 lists.

Figure 8: Summary of technical filtering for the methylation dataset. **(a)** Overlap between CpGs and the Naeem, Pidsley, Zhou, Chen, and McCartney lists. **(b)** Redundancy analysis quantifying concordance among the technical annotations.



(a) r_β distribution on Normal+Adjacent samples, with thresholds at 0.01, 0.02, and 0.05.

(b) r_β distribution on Normal+Tumor samples under the same thresholds.

Figure 9: Variability range analysis for GSE225845. A large block of CpGs shows low dispersion in both N+A and N+T groups, supporting the use of invariance filtering to remove biologically uninformative loci.

Step 3 – Invariance Filtering on Normal+Adjacent For each CpG, the variability statistic $r_\beta = P90(\beta) - P10(\beta)$ was computed on the ✓ 530,683 autosomal probes remaining after technical filtering. The distribution displayed a substantial concentration of loci with limited dynamic range, supporting the application of invariance filtering.

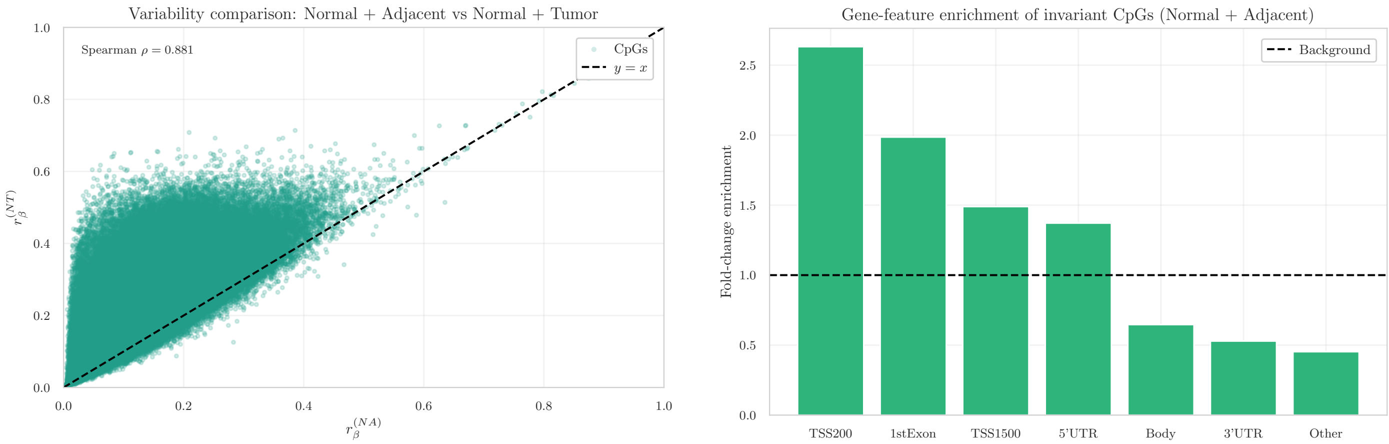
Using the study threshold $r_\beta < 0.05$, a total of ✓ 130,939 CpGs (✓ 24.7%) were classified as non-variable in N+A tissue and removed. The resulting feature matrix `beta_final_preFS` contains ✓ 399,744 CpGs across ✓ 477 samples.

To evaluate the stability of variability estimates when tumour samples are included, r_β was computed on the N+T subset, identifying ✓ 103,622 non-variable CpGs. The comparison between N+A and N+T revealed strong concordance (✓ Spearman $\rho = 0.881$), with a visible deviation band corresponding to CpGs that lose stability when tumour samples are added. Set comparisons yielded:

- **Stable invariants** (N+A ∩ N+T): ✓ 102,765 CpGs;
- **Breakdown loci** (N+A \ N+T): ✓ 28,174 CpGs;
- **N+T-only invariants**: ✓ 857 CpGs.

The invariants were compared with the universal CpGs reported by Edgar (✓ 42,315 loci shared across blood, buccal, and placenta tissues). In this dataset:

- ✓ 27,895 were invariant in N+A breast tissue;



(a) Comparison of N+A vs. N+T variability. Breakdown loci deviate from the identity line.

(b) Genomic enrichment of N+A invariant CpGs across annotated gene features.

Figure 10: Variability concordance and genomic context of invariant CpGs. Promoter-associated regions show the strongest enrichment, while tumour-induced breakdown loci reveal loss of stability upon inclusion of tumour samples.

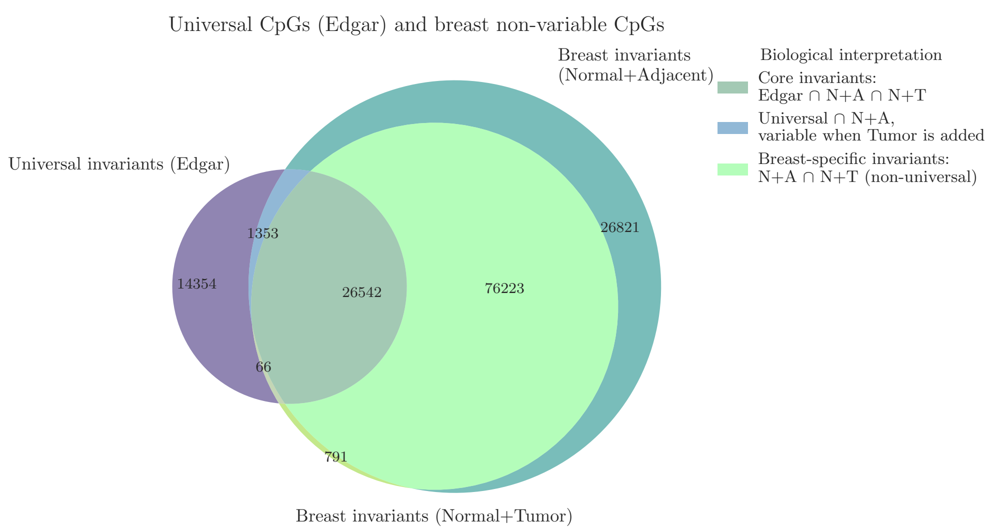
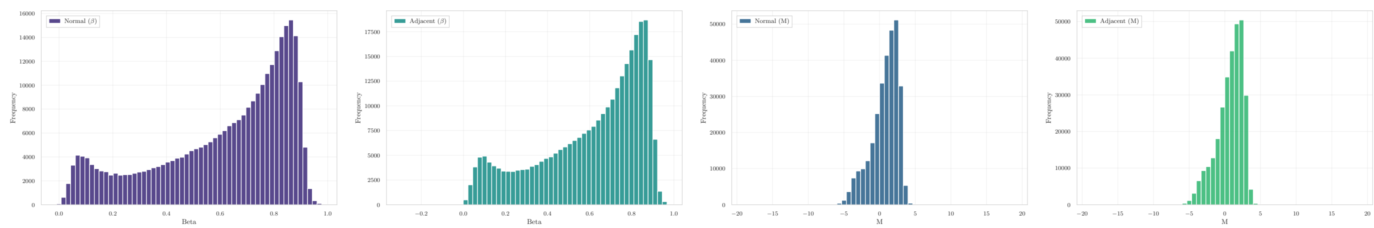


Figure 11: Intersection between universal invariant CpGs (Edgar) and breast-tissue invariants. Breakdown loci represent sites with conserved stability across tissues that lose invariance in the tumour context.

- ✓ 26,608 remained invariant in N+T;

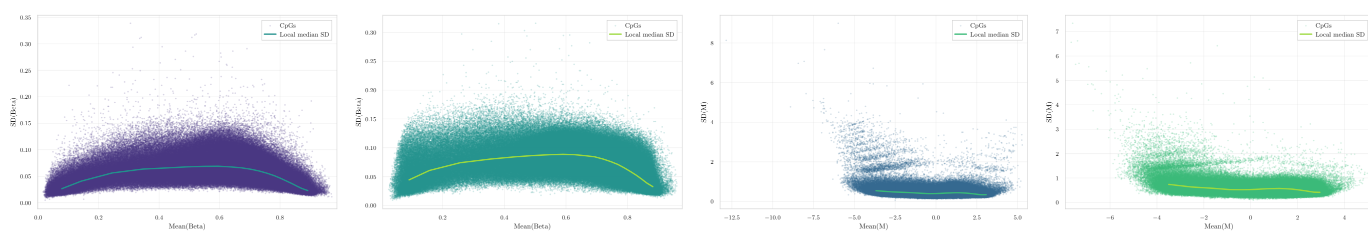
- ✓ 1,353 universal loci became variable in tumour tissue (breakdown).

Applying the threshold $r_\beta < 0.05$ yielded ✓ 130,939 N+A invariants and ✓ 399,744 retained CpGs, now forming the working autosomal matrix for downstream modelling. All per-probe decisions have been recorded in the updated removal summary, which now includes an Edgar invariance flag and an updated `Source_Count`.



(a) Normal (β) (b) Adjacent (β) (c) Normal (M) (d) Adjacent (M)

Figure 12: **Distributional comparison between β and M quantifications in Normal and Adjacent breast tissue.** In β-space, both groups display the expected U-shaped distribution, whereas the logit transformation yields approximately symmetric patterns in M-space. All panels computed on the filtered set of ✓ 399,744 CpGs.



(a) Mean–SD Normal (β) (b) Mean–SD Adjacent (β) (c) Mean–SD Normal (M) (d) Mean–SD Adjacent (M)

Figure 13: **Mean–SD relationship before and after $\beta \to M$ conversion.** The pronounced mean-dependent variance in β-space is substantially flattened in M-space, confirming the variance-stabilising effect of the logit transformation.

Step 4 – β to M Conversion Following technical and invariance filtering, the working matrix contained ✓ 399,744 CpGs across ✓ 477 samples. All β-values were transformed into M-values using the standard logit relation yielding an unbounded and approximately homoscedastic scale suitable for linear modelling and batch correction procedures. The transformed dataset, denoted `M_clean`, preserves the same sample structure and CpG ordering as the β matrix and was saved in Parquet format for downstream analysis.

A manual inspection of numerical values confirmed correct transformation across the entire CpG panel, including loci with extreme methylation levels.

Distributional effects of the transformation. The empirical distributions of Normal and Adjacent samples, shown in Fig. 12, exhibit the characteristic U-shape in β-space and the expected approximately symmetric profiles after conversion. The mean–SD relationship demonstrates the well-known variance inflation at intermediate β levels, which becomes substantially stabilised in M-space (Fig. 13). These patterns confirm that the M-value representation provides a more appropriate scale for statistical inference.

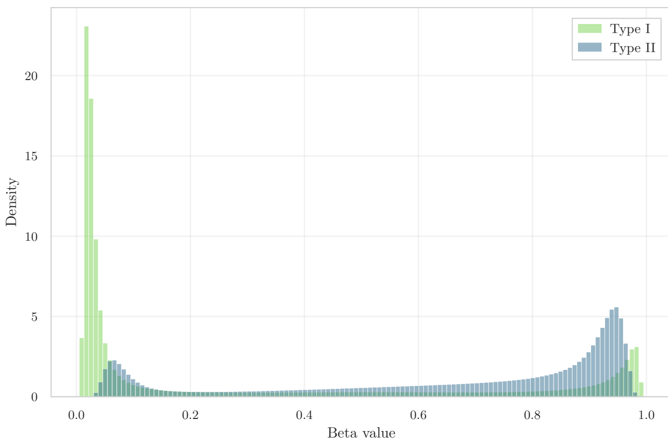
The filtered β-matrix (✓ 399,744 CpGs, ✓ 477 samples) was successfully converted into the corresponding M-value representation and stored as a Parquet object. M-values will be used for all inferential analyses, while β-values will be retained for visualisation and biological interpretation.

Step 5 – Batch Effect Correction

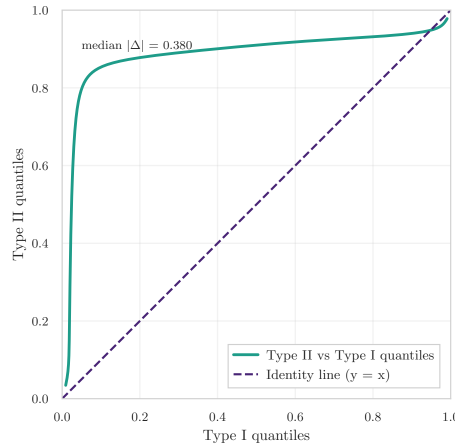
Step 6 – Final Outputs

1.3 Dataset GSE287331

Step 0 – Initial Pre-processing The GSE287331 dataset was imported from its Parquet representation, yielding an initial β-value matrix of ✓ 446 samples and ✓ 703,168 CpGs. The accompanying phenotype table reported the same

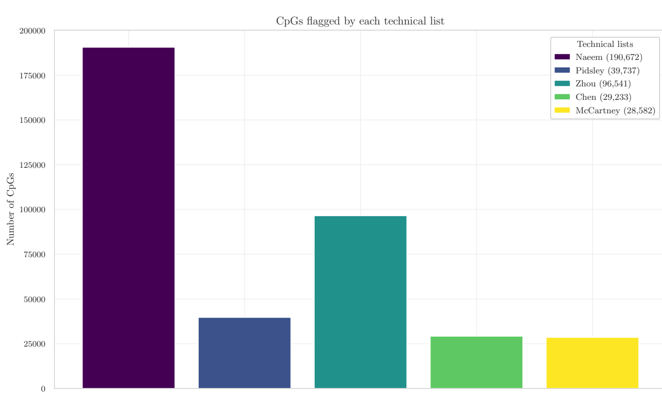


(a) Type I vs. Type II density (pre BMIQ).

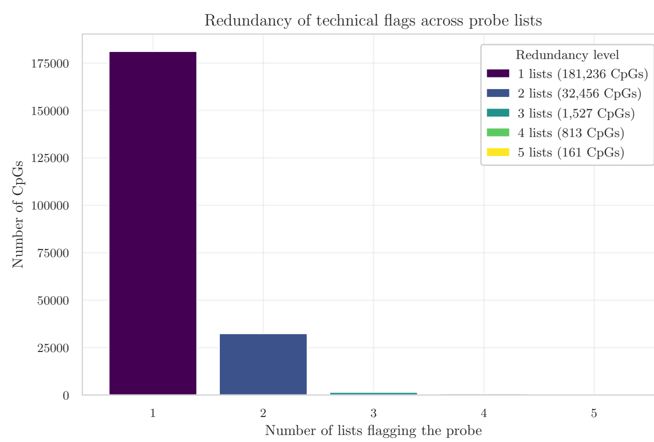


(b) Q–Q comparison of Type II vs. Type I probes (pre BMIQ). The median absolute deviation of ✓ 0.380 indicates a strong quantile-wise shift, confirming the absence of probe-type normalization in the original dataset.

Figure 14: Diagnostic evaluation of Infinium probe–design bias before and after BMIQ. The correction reduces the discrepancy only marginally, and substantial distributional differences remain between probe types.

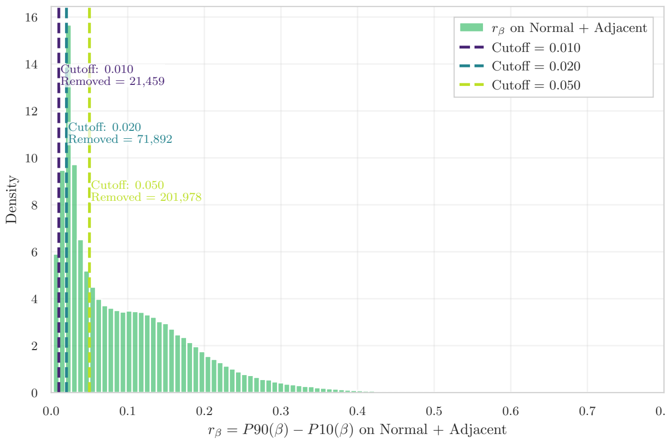


(a) CpGs flagged by each technical list.

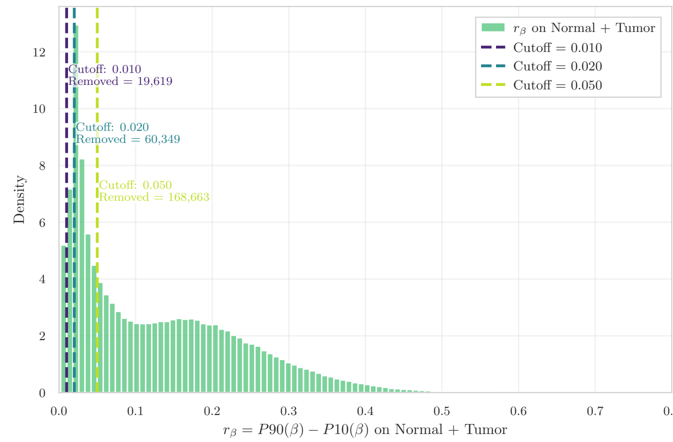


(b) Redundancy of probes flagged by 1–5 lists.

Figure 15: Summary of technical filtering for the methylation dataset. **(a)** Number of CpGs intersecting each curated resource. **(b)** Redundancy analysis capturing concordance across filtering lists.

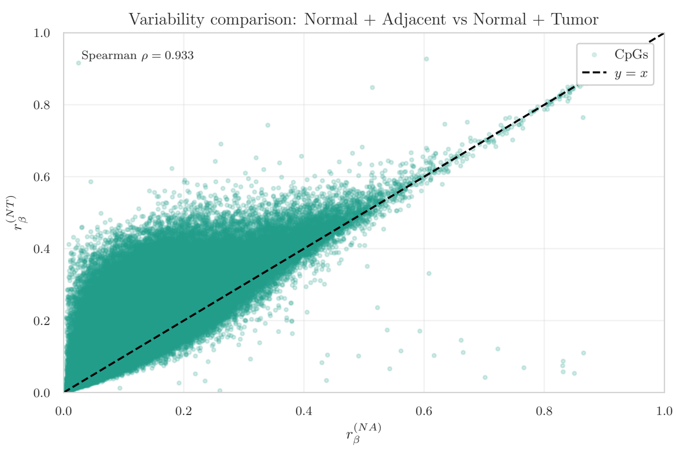


(a) r_β distribution on Normal+Adjacent samples.

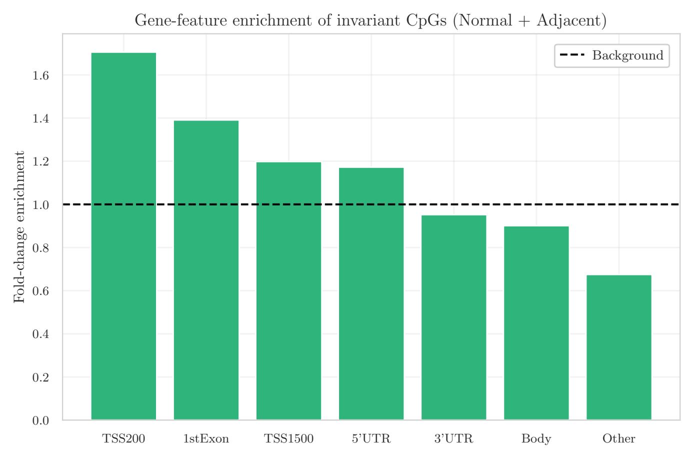


(b) r_β distribution on Normal+Tumor samples.

Figure 16: Variability range distributions for GSE287331. A substantial block of CpGs displays low dispersion in both N+A and N+T subsets, supporting the application of invariance filtering.



(a) Comparison of N+A vs. N+T variability.



(b) Genomic enrichment of N+A invariant CpGs.

Figure 17: Variability concordance and genomic context of invariant CpGs. Promoter-associated elements exhibit the strongest enrichment.

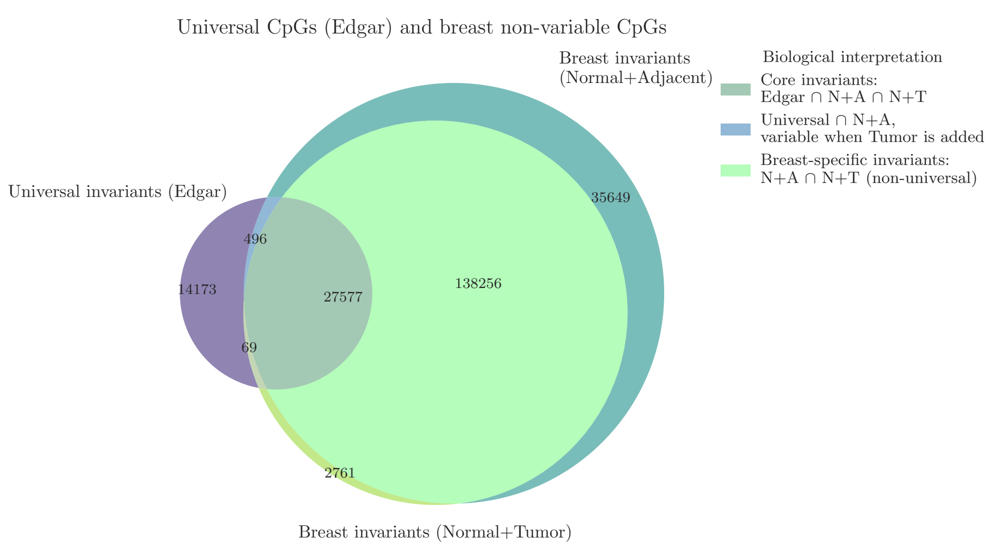
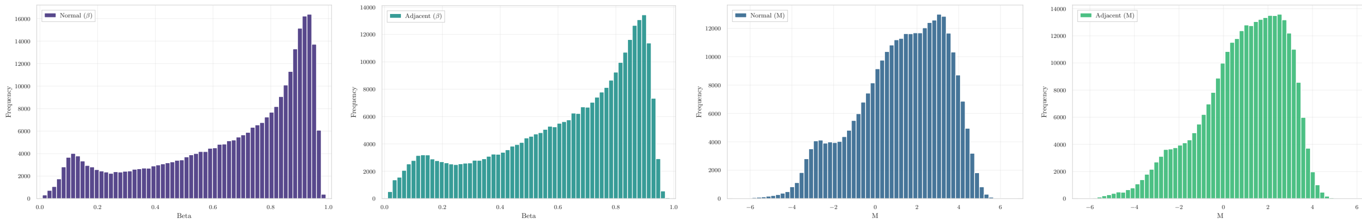
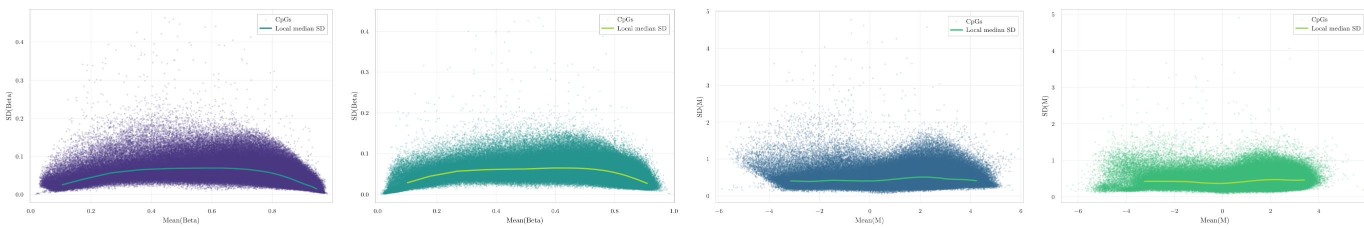


Figure 18: Intersection between universal invariants (Edgar) and breast-tissue invariants. Breakdown loci represent universal CpGs that lose stability specifically in tumour tissue.



(a) Normal (β) (b) Adjacent (β) (c) Normal (M) (d) Adjacent (M)

Figure 19: **Empirical distributions of β-values and M-values.** The U-shaped β distributions become substantially more symmetric after logit transformation, without distorting relative methylation patterns between Normal and Adjacent tissue.



(a) Mean–SD (Normal, β) (b) Mean–SD (Adjacent, β) (c) Mean–SD (Normal, M) (d) Mean–SD (Adjacent, M)

Figure 20: **Mean–SD structure before and after $\beta \to M$ transformation.** M-values exhibit markedly reduced dependence between mean and variance, providing a more suitable scale for downstream statistical inference.

number of entries and contained all required metadata fields. Before continuing with probe-level preprocessing, the dataset underwent a dedicated missingness-cleaning stage to ensure high-quality input for downstream analysis.

Following phenotype harmonisation and sample selection criteria specific to this dataset, the working cohort consisted of ✓ 311 samples, which were used for all subsequent steps of the pipeline. Sample identifiers were unique, correctly formatted, and in perfect correspondence between the methylation and phenotype tables.

Missingness filtering and imputation Probe-wise missingness was quantified across the β-matrix. Consistent with recommendations from Islam *et al.* [18] and Mansell *et al.* [19], a stringent threshold was applied: all CpG sites missing in more than ✓ 1% of samples were removed. This filter ensures high-confidence measurements while leveraging the large dimensionality of the EPIC platform.

After applying this criterion, the remaining level of sparsity was extremely low, with only ✓ 58,561 NaN values (✓ 0.0187% of all entries). These residual missing values were imputed using probe-wise median substitution, a strategy that avoids introducing group-specific bias and is also adopted in recent harmonisation workflows such as mLiftOver [20].

The resulting β-matrix contained no missing values and was suitable for probe-design diagnostics and technical filtering.

Data integrity verification After missingness filtering and imputation, the final working matrix consisted of ✓ 311 samples and ✓ 703,168 CpGs. All entries were fully observed (✓ 0 residual NaN), and sample alignment between metadata and methylation values was complete, allowing the preprocessing pipeline to proceed with Infinium Type I/II bias diagnostics.

Step 1 – Infinium Type I/II Bias Diagnostics A probe–design diagnostic analysis was performed to assess potential discrepancies between Infinium Type I and Type II probes. Using the official EPIC v1.0 manifest, ✓ 115,705 Type I and ✓ 585,514 Type II probes were identified.

The β-value distributions of the two chemistries showed a marked mismatch (Fig. 14a), with Type II probes displaying a strong right-shifted profile. The quantile–quantile comparison confirmed a substantial systematic offset, with a median absolute deviation of ✓ 0.380 (Fig. 14b), indicating the absence of any Infinium probe-design bias correction in the originally released data.

Given this discrepancy, a per-sample BMIQ normalization step was applied using the `wateRmelon` implementation, operating strictly on individual samples to minimise cross-sample distortion. Post-correction diagnostics showed a modest reduction in probe-type mismatch ($|\Delta| : 0.380 \to 0.370$) and a slight improvement in the alignment of high-β regions, but the overall distributional discrepancy remained substantial. Due to this limited harmonization and the known tendency of BMIQ to attenuate biological differences when global distributional shifts are present, the corrected matrix is not used for downstream modelling. All subsequent analyses therefore rely on the non-BMIQ version of the dataset, which preserves clearer group-level structure and avoids normalization-induced shrinkage.

Step 2 – Technical Filtering Technical probe filtering was applied to exclude CpG loci affected by unreliable hybridisation, SNP interference, multi-mapping, or annotation inconsistencies. The initial working matrix contained ✓ 702,918 CpGs across the 311 selected samples. The intersection with established technical probe-filtering resources showed substantial overlap across multiple lists.

- ✓ 124,511 CpGs flagged by Naeem et al. [10];
- ✓ 8,992 CpGs in the Pidsley/Chen blacklist [7], [8];
- ✓ 102,802 CpGs annotated by Zhou's `MASK_*` fields [6];
- ✓ 1,730 cross-reactive probes from Chen et al. [8];
- ✓ 3,672 cross-hybridising loci identified by McCartney et al. [9].

The union of all lists resulted in the removal of ✓ 203,114 unique CpGs, leaving ✓ 499,804 CpGs before annotation-based filtering.

Annotation-based filtering (Illumina manifest) Using the EPIC v1.0 manifest, probes lacking valid genomic coordinates or targeting non-CpG contexts (IDs beginning with "`ch`") were removed. This eliminated ✓ 500 non-CpG probes, reducing the feature set to ✓ 499,552 CpGs.

Removal of sex-chromosome probes (X/Y) Probes mapped to chromosomes X and Y were removed to avoid sex-driven confounding. A total of ✓ 12,579 CpGs were discarded, resulting in an autosomal working set of ✓ 486,973 CpGs.

Technical filtering summary Across all filters, ✓ 216,193 unique CpGs were removed from the initial ✓ 702,918, leaving a final autosomal set of ✓ 486,725 CpGs. A probe-level removal summary, including all contributing lists and the `Source_Count` redundancy indicator, is provided in `removed_cpgs_all_filters_summary.csv`.

Step 3 – Invariance Filtering on Normal+Adjacent A variability-based filtering analysis was performed to identify CpG loci exhibiting minimal dispersion in Normal+Adjacent (N+A) breast tissue. For each CpG, the variability statistic

$$r_\beta = P90(\beta) - P10(\beta)$$

was computed across ✓ 486,725 autosomal probes, producing the distribution shown in Fig. 16. The density exhibited a large concentration of low-variability loci. Applying the study threshold $r_\beta < 0.05$ resulted in ✓ 201,978 non-variable CpGs (✓ 41.5%), leaving ✓ 284,747 CpGs for downstream modelling.

Variability comparison with Normal+Tumor To assess whether tissue-specific variability patterns were preserved upon inclusion of tumour samples, the same computation was applied to the N+T subset, identifying ✓ 168,663 non-variable CpGs. The two variability measures showed strong concordance (✓ Spearman $\rho = 0.933$), with a distinct set of loci deviating from the identity line (Fig. 17). These correspond to CpGs that lose stability when tumour samples are added.

The set comparison yielded:

- **Stable invariants** (N+A $\cap$ N+T): ✓ 165,833 CpGs;
- **Breakdown loci** (N+A $\setminus$ N+T): ✓ 36,145 CpGs;
- **N+T-only invariants**: ✓ 2,830 CpGs.

Comparison with universal invariant CpGs (Edgar et al.) Invariant loci were compared with the universal CpG set reported by Edgar (✓ 42,315 CpGs). In this dataset:

- ✓ 28,073 overlapped with N+A invariants;
- ✓ 27,646 remained invariant in N+T;
- ✓ 496 universal invariants became variable when tumour samples were added.

Using the threshold $r_\beta < 0.05$, a total of ✓ 201,978 N+A invariants (✓ 41.5%) were removed, yielding a working matrix of ✓ 284,747 CpGs across ✓ 446 samples. All decisions were recorded in the updated per-probe summary, which now includes an Edgar invariance flag and updated `Source_Count`.

Step 4 – β to M Conversion Given the strong mean–variance dependence intrinsic to β-values, the working matrix obtained after technical and invariance filtering (✓ 284,747 CpGs across ✓ 446 samples) was transformed into M-values using the logit mapping

$$M = \log_2\left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon}\right), \qquad \varepsilon = 10^{-6}.$$

This transformation yields an approximately homoscedastic scale and is the standard input for linear models, empirical Bayes moderation, and batch-effect correction.

The transformation was applied to all CpG columns (✓ 284,745 CpGs, excluding `id_tissue` and `label`). The resulting object, `M_clean`, preserves the same schema as the β-matrix and was stored as a memory-efficient `LazyFrame`.

As in previous datasets, M-values were computed only for samples belonging to the Normal, Adjacent, and Tumor groups. Since the harmonized phenotype indicates that ✓ 311 samples belong to these categories (labels 0/1/2), the final M-matrix contains:

M_clean shape: 3 311 $\times$ 284,747.

This reduction reflects the exclusion of samples annotated with non-biological labels (e.g. quality-control entries or non-breast tissues), and is consistent with the dataset's experimental design.

A preview of the resulting table confirms numerical correctness across the full CpG panel, including loci with extreme β values near the boundaries of the unit interval.

Distributional effects of the $\beta \to M$ transformation. Figures 19 and 20 illustrate the impact of the transformation on the empirical data distribution. As expected, the U-shaped β-profiles for Normal and Adjacent tissues become substantially more symmetric after conversion, while the pronounced mean–SD dependence flattens into a markedly more stable variance structure across the full range of M-values. These properties make M-values preferable for statistical modelling, whereas β-values are retained for biological interpretation and visualisation.

Summary The M-value matrix (✓ 311 samples × ✓ 284,747 CpGs) was saved to disk in compressed Parquet format and is used for all modelling tasks. The β-matrix is retained for interpretability-focused analyses such as heatmaps, density plots, and outlier visualisation.

2 Inter-Dataset Analysis

A Filtering lists

This appendix details the major technical filtering lists and annotations applied in Illumina 450K and EPIC methylation arrays to remove unreliable or ambiguous CpG probes.

A.1 Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the humanmethylation450 array

Naeem *et al.* [10] proposed a structured filtering framework to reduce false discoveries by sequentially discarding probes based on hybridisation specificity and genomic integrity (Figure 21). The main exclusion steps are:

- Multi-mapping or cross-hybridising probes ⇒ *discard*.
- Probes overlapping repetitive elements (LINE/SINE/ALU) ⇒ *discard*.
- Probes targeting regions with INDELs ⇒ *discard*.
- Probes overlapping SNPs:
  - SNP at CpG or extension site ⇒ *discard*.
  - SNP near target but not interfering in bisulfite space ⇒ *keep*.

The resulting "discard" set therefore removes probes affected by cross-reactivity, structural polymorphisms (INDELs), and SNPs that alter CpG interrogation.

A.2 Discovery of cross-reactive probes and polymorphic cpgs in the illumina infinium humanmethylation450 microarray

Chen *et al.* [8] empirically identified probes in the 450K array that exhibit off-target hybridisation or overlap with common SNPs. For this project, only the **cross-reactive list** is available and used. This list enumerates ∼29k multi-mapping probes whose methylation signal cannot be uniquely attributed to one genomic locus.

A.3 Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling

Pidsley *et al.* [7] provide a platform-level assessment of the EPIC array, with explicit *annotated probe lists* that flag probes whose methylation signal may be confounded by design- or genome-related artefacts. The focus is on probe categories and positions where technical bias arises, rather than on a universal, prescriptive blacklist. In particular, they catalogue:

- **Cross-hybridising CpG-targeting probes:** CpG probes showing sequence homology (off-target matches) to additional genomic loci, yielding non-unique hybridisation and potentially inflated or ambiguous β signals.
- **Cross-hybridising non-CpG-targeting probes:** off-target issues among CNG/non-CpG probes; these are typically excluded in CpG-centric analyses due to limited interpretability and higher risk of artefacts.
- **Probes overlapping common genetic variation:** annotation of variants from population data at three critical positions:
  - *At the interrogated CpG* (polymorphic CpG) — directly affects the presence of the CpG dinucleotide and the measured methylation state;
  - *At the single-base extension (SBE) site* (Type I) — perturbs extension and dye chemistry, biasing intensity ratios;
  - *Within the probe body* — can reduce binding affinity or alter hybridisation kinetics, especially for common variants.

Table 1: Technical categories covered by each filtering resource.

Category	Naeem A.1	Chen A.2	Pidsley A.3	Zhou A.4	McCartney A.5
Cross-hybridisation / multi-mapping	Yes (Steps 1–2)	Yes	Yes	`MASK_mapping`	Table 2+3
SNP at CpG / extension base	Yes (Step 5)	–	Yes	`MASK_snp5, MASK_extBase`	–
Nearby SNP tolerated	Conditional (Step 6)	–	–	–	–
INDEL / structural variant	Yes (Step 3)	–	–	–	–
Non-CpG probes	–	–	Yes	`MASK_nonCG`	Table 3

A.4 Comprehensive characterization, annotation and innovative use of infinium dna methylation beadchip probes

Zhou *et al.* [6] released a unified probe annotation for both 450K and EPIC arrays with multiple binary mask columns (`MASK_*`). Each mask flags probes to be removed due to specific technical artifacts.

- `MASK.mapping`: probes with low or inconsistent mapping quality (non-unique alignment or presence of INDELs);
- `MASK.typeINextBaseSwitch`: Type-I probes carrying a SNP in the extension base that causes a color-channel switch (*CCS* probes);
- `MASK.extBase`: probes whose extension base is inconsistent with the expected color channel or CpG context;
- `MASK.sub30.copy`: probes with non-unique 30-bp 3' subsequences (potential cross-hybridisation);
- `MASK.snp5.common`: probes overlapping a SNP within ±5 bp of the interrogated CpG (even with global MAF ¡ 1%);
- `MASK.snp5.GMAF1p`: probes overlapping SNPs with global MAF ¿ 1%;
- `MASK.general`: recommended composite mask integrating mapping, SNP, and cross-reactivity filters for general use;
- `MASK.rmsk15`: probes overlapping RepeatMasker regions (not recommended for exclusion in standard workflows).

This resource provides a programmatic and reproducible way to perform fine-grained probe masking.

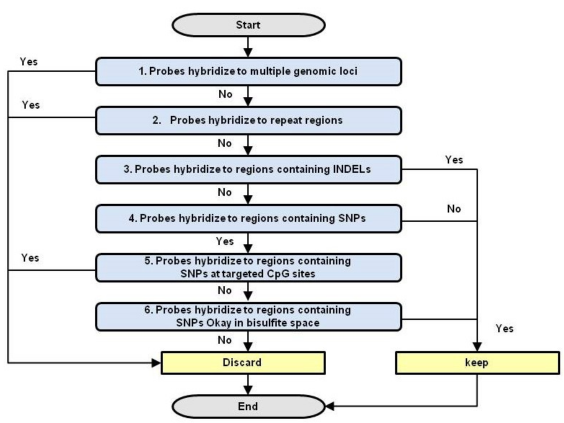


Figure 21: Workflow for determining affected Probes.

A.5 Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip

McCartney *et al.* [9] assessed probe design artifacts and published supplementary tables that include:

- Cross-hybridising CpG-targeting probes;
- Cross-hybridising non-CpG-targeting probes.

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